

MIAO LU

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EDUCATION

Stanford University

Ph.D. Candidate in Operations Research

Advisor: Prof. Jose Blanchet

Stanford, USA

Sep.2023 – present

University of Science and Technology of China

B.S. in Mathematics & Applied Mathematics

Ranking: 2/140, with summa cum laude (Guo Moruo Scholarship)

Hefei, China

Sep.2018 – Jun.2022

INDUSTRIAL EXPERIENCES

NVIDIA Research, Research Scientist Intern

Host: Dr. Yaosheng Fu and Dr. Zhiding Yu

Research on LLM agent long-context training data synthesis.

Santa Clara, USA

Feb.2026 – present

ByteDance Seed, Research Scientist Intern

Host: Dr. Jiecao Chen

Developed end-to-end agentic RL algorithms to scale LLM agent training for long horizon tasks beyond a fixed context limit, with techniques of summarization and context-folding driven context management. Achieved significant improvements over RL baseline in deep research and agentic coding tasks. Progress results in publications [C11][C13] (ACL'26, ICML'26).

San Jose, USA

Jun.2025 – Dec.2025

Ubiquant Investment, Quantitative Research Intern

Interned at AI department of Ubiquant, research on DL and RL for quantitative trading.

Shanghai, China

Jun.2022 – Sep.2022

RESEARCH EXPERIENCES

Stanford University, Graduate Research Assistant

Advisor: Prof. Jose Blanchet

Projects: (i) LLM RLHF from limited data and distributional shifts, the first to propose SFT as regularization in preference learning with theoretical grounding [C9]; (ii) theory and algorithm of distributionally robust decision-making [C6][C8][P2].

Stanford, USA

Sep.2023 – present

Toyota Technological Institute at Chicago, Student Visitor

Host: Prof. Tianhao Wang and Prof. Zhiyuan Li

Projects: (i) theoretical understanding of heavy-ball momentum acceleration with large learning rates in river-valley loss landscapes [P3]; (ii) optimal computational-statistical trade-off of learning single-index models via neural networks [C10].

Chicago, USA

July.2024 – Sep.2024

Northwestern University, Remote Research Assistant

Host: Prof. Zhaoran Wang

Selected projects: Principled exploration for online RL under function approximations via value-incentivized regularization (Maximize to Explore [C5]). The method inspires a long line of following work in online/offline RL, multi-agent RL, RLHF.

Remote

Sep.2021 – Aug.2023

INVITED TALKS

Scaling Long Horizon LLM Agent via Reinforcement Learning [C11][C13]

- o TikTok Research Seminar, remote [P4][P5]

Oct.2025

Theoretical Foundations of Distributionally Robust Decision Making [C6][C8][P2]

- o 2025 INFORMS annual meeting, Atlanta GA, USA [P1][P2]
- o 2024 INFORMS annual meeting, Seattle, WA, USA [C8]
- o 58th Annual Conference on Information Sciences and Systems (CISS), Princeton, NJ, USA [C6]
- o 2023 INFORMS annual meeting, Phoenix, AZ, USA [C6]

Oct.2025

Oct.2024

Mar.2024

Oct.2023

Computational-statistical Trade-off of Learning Single-index Models via Neural Networks [C10]

- o 2nd Mathematics of Modern Machine Learning Workshop (M3L), Vancouver, BC, Canada

Dec.2024

Authors with * contributed equally to the work, and [†] represents alphabetical order.

- [P4] **INFUSER: Influence-Guided Self-Evolution Improves Reasoning**
Siyu Chen, **Miao Lu**, Beining Wu, Heejune Sheen, Fengzhuo Zhang, Shuangning Li, Zhiyuan Li, Jose Blanchet, Tianhao Wang, Zhuoran Yang
Preprint under conference review
- [P3] **Towards Understanding Momentum Acceleration in River-Valley Loss Landscape**
Miao Lu, Zeyu Bian, Kaiyue Wen, Beining Wu, Siyu Chen, Tianhao Wang, Zhiyuan Li
Preprint under conference review
- [P2] **Robust Assortment Optimization from Observational Data**
Miao Lu, Yuxuan Han, Han Zhong, Zhengyuan Zhou, Jose Blanchet
arXiv preprint
- [P1] **Learning an Optimal Assortment Policy under Observational Data**
Yuxuan Han, Han Zhong, **Miao Lu**, Jose Blanchet, Zhengyuan Zhou
Preprint under review at Management Science (MS)
- [C13] **Scaling Long-Horizon LLM Agent via Context Folding**
Weiwei Sun, **Miao Lu**, Zhan Ling, Xuesong Yao, Kang Liu, Yiming Yang, Jiecao Chen
International Conference on Machine Learning (ICML) 2026
- [C12] **Towards Theoretical Understanding of Transformer Test-Time Computing: Investigation on In-Context Linear Regression**
Xingwu Chen, **Miao Lu**^{*}, Beining Wu, Difan Zou
NeurIPS Workshop on Foundations of Reasoning in Language Models (FoRLM) 2025
International Conference on Machine Learning (ICML) 2026
- [C11] **Beyond the Context Window: Scaling Agentic RL via End-to-end Optimized Context Compression**
Miao Lu, Weiwei Sun, Weihua Du, Zhan Ling, Xuesong Yao, Kang Liu, Jiecao Chen
Association for Computational Linguistics (ACL) 2026
- [C10] **Can Neural Networks Achieve Optimal Computational-statistical Tradeoff? An Analysis on Single-Index Model**
Siyu Chen^{*}, Beining Wu^{*}, **Miao Lu**, Zhuoran Yang, Tianhao Wang
International Conference on Learning Representations (ICLR) 2025 **Oral presentation**
NeurIPS Workshop on Mathematics of Modern Machine Learning (M3L) 2024 **Oral presentation**
- [C9] **Provably Mitigating Overoptimization in RLHF: Your SFT Loss is Implicitly an Adversarial Regularizer**
Zhihan Liu^{*}, **Miao Lu**^{*}, Shenao Zhang, Boyi Liu, Hongyi Guo, Yingxiang Yang, Jose Blanchet, Zhaoran Wang
Neural Information Processing Systems (NeurIPS) 2024
ICML Workshop on Aligning Reinforcement Learning Experimentalists and Theorists (ARLET) 2024
- [C8] **Distributionally Robust Reinforcement Learning with Interactive Data Collection: Fundamental Hardness and Near-Optimal Algorithm**
Miao Lu^{*}, Han Zhong^{*}, Tong Zhang, Jose Blanchet
Neural Information Processing Systems (NeurIPS) 2024
ICML Workshop on Aligning Reinforcement Learning Experimentalists and Theorists (ARLET) 2024
- [C7] **Benign Oscillation of Stochastic Gradient Descent with Large Learning Rates**
Miao Lu^{*}, Beining Wu^{*}, Xiaodong Yang, Difan Zou
International Conference on Learning Representations (ICLR) 2024
NeurIPS Workshop on Mathematics of Modern Machine Learning (M3L) 2023
- [C6] **Double Pessimism is Provably Efficient for Distributionally Robust Offline Reinforcement Learning: Generic Algorithm and Robust Partial Coverage**
Jose Blanchet[†], **Miao Lu**[†], Tong Zhang[†], Han Zhong[†]
Neural Information Processing Systems (NeurIPS) 2023
Extended version under major revision at Mathematics of Operations Research (MOR)
- [C5] **Maximize to Explore: One Objective Function Fusing Estimation, Planning, and Exploration**
Zhihan Liu^{*}, **Miao Lu**^{*}, Wei Xiong^{*}, Han Zhong, Hao Hu, Shenao Zhang, Sirui Zheng, Zhuoran Yang, Zhaoran Wang
Neural Information Processing Systems (NeurIPS) 2023 **Spotlight**
Extended version under review at Operations Research (OR)
- [C4] **Pessimism in the Face of Confounders: Provably Efficient Offline Reinforcement Learning in Partially Observable Markov Decision Processes**

Miao Lu, Yifei Min, Zhaoran Wang, Zhuoran Yang
International Conference on Learning Representations (ICLR) 2023

[C3] **Welfare Maximization in Competitive Equilibrium: Reinforcement Learning for Markov Exchange Economy**
Zhihan Liu*, **Miao Lu***, Zhaoran Wang, Michael I. Jordan, Zhuoran Yang
International Conference on Machine Learning (ICML) 2022

[C2] **Learning Pruning-Friendly Networks via Frank-Wolfe: One-Shot, Any-Sparsity, and No Retraining**
Miao Lu*, Xiaolong Luo*, Tianlong Chen, Wuyang Chen, Dong Liu, Zhangyang Wang
International Conference on Learning Representations (ICLR) 2022 **Spotlight**

[C1] **Learning Robust Policy against Disturbance in Transition Dynamics via State-Conservative Policy Optimization**
Yufei Kuang, **Miao Lu**, Jie Wang, Qi Zhou, Bin Li, Houqiang Li
Association for Advancement of Artificial Intelligence (AAAI) 2022

AWARDS AND HONORS

Xinhe Scholarship (outstanding undergraduate researchers, School of the Gifted Young, USTC)	<i>Mar.2023</i>
Yuanqing Yang Scholarship (top scholarship, School of Mathematical Sciences, USTC)	<i>Jan.2022</i>
The 41st Guo Moruo Scholarship (highest honor, USTC)	<i>Dec.2021</i>
Chinese National Scholarship (top scholarship, Ministry of Education of China)	<i>Nov.2019, 2020</i>

ACADEMIC SERVICES

Journal Reviewer: Annals of Applied Probability (AOAP), Operations Research (OR), Management Science (MS), Mathematics of Operations Research (MOR), Transactions on Machine Learning Research (TMLR)

Conference Reviewer: Neural Information Processing Systems (NeurIPS; 2023-2025), International Conference on Machine Learning (ICML; 2024-2026), International Conference on Learning Representations (ICLR; 2024-2026), International Conference on Artificial Intelligence and Statistics (AISTATS; 2025), ICML Workshop on Aligning Reinforcement Learning Experimentalists and Theorists (ARLET; 2024), ICML Workshop on Exploration in AI Today (EXAIT; 2025), ICML Workshop on Methods and Opportunities at Small Scale (MOSS; 2025), NeurIPS Workshop on Mathematics of Modern Machine Learning (M3L; 2024), Association for the Advancement of Artificial Intelligence (AAAI; 2025)